Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/17/2011
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Antigen-Specific CD4+ T Cell Priming and Memory Response During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	milliehf1@gmail.com	**Fax:**	FY 415-750-6667
PI Organization Type:	NON-PROFIT	**Phone:**	415-750-6940
Organization Name:	Northern California Institute for Research and Education		
PI Address 1:	Lab of Cell Growth, Mail Code 151F, VAMC		
PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	04/15/2009	**End Date:**	04/13/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 ; ISS NOTE: New end date is 4/13/2013 per A. Chu/ARC (Ed., 1/27/2012) NOTE: New end date is 4/14/2012 per A. Chu/ARC (Ed., 3/10/2011) NOTE: Grant number corrected to NNX09AH21G (from NNX09AG21G) per A. Chu/ARC (4/2010) NOTE: End date changed to 4/14/2011 (from 4/14/2012) per A. Chu/ARC (7/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chang, Tammy (Veterans Affairs Medical Center)		
Grant/Contract No.:	NNX09AH21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight immunosuppression is a significant obstacle to long-term human space travel. Of foremost concern is whether astronauts/cosmonauts may be able to generate effective protective immune responses against infections while in spaceflight. Using an innovative mouse experimental model, we propose a set of experiments that will test 1) whether initial specific activation of T cells is intact and 2) whether memory T cell function is maintained during spaceflight. Memory T cells are the mediators of immune protection after vaccinations and whether they function properly during spaceflight will determine if pre-flight vaccinations may be useful interventions. In our experimental system, transgenic T cells specific for a model protein will be transferred into recipient mice that have implanted mini-pumps. The mini-pumps will release the model protein along with non-infectious purified bacterial cell wall components. Specific T cells stimulated under these inflammatory conditions will be similar to those activated in a bacterial infection but no infectious agents will be involved. Mini-pumps will delay T cell stimulation until after mice acclimate to spaceflight to ensure that initiation of T cell activation occurs in microgravity. Naïve (previously unactivated) T cells will be transferred to determine whether responses to new immunological challenges (e.g. new infections) are intact. Previously activated T cells that will develop into memory cells will be transferred to determine whether robust secondary responses may be elicited during spaceflight. Our proposal directly addresses the research priorities of the Bion-M1 mission (changed to a Space Shuttle mission in 2010) to investigate resistance to infections and acquired immune responses without using infectious agents onboard the spacecraft. Together, these proposed experiments will address some of the most pressing outstanding questions in spaceflight immunology and more clearly define the risks and potential countermeasures of spaceflight immunosuppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The alteration of the immune system in spaceflight has been suggested by some to have many analogies to changes in the immune system during aging. In these studies of the immune system in mice, we have found that the changes in the immune function in flown mice are much the same as in humans. This suggests that there are common mechanism of spaceflight immunosuppression in both mammals. Discoveries made from this study will help us understand the immune system and the basic initial steps of activation as well as help find new targets to use in the treatment of immunosuppression observed in aging.
Task Progress:	The objective of this study was to evaluate the effect of spaceflight on early T cell activation. C57BL/6J female mice were exposed to microgravity for 15 days on STS-131. Splenocytes were harvested within 2-3 hrs post-flight, cells isolated and activated (~9 hrs after landing) with Concanavalin A and antiCD28 for 2.5 hrs. Cells were harvested (11.5 hrs post landing) and stored in RNAprotect (Qiagen) at -80°C. Total RNA was isolated from the activated samples with RNeasy (Qiagen). The flight was successful and the animals were brought back in great condition. The progress of this experiment is proceeding well. Analysis of Aim I (Aim I. Determine whether antigen-specific CD4+ T cell priming is inhibited or dysfunctional after spaceflight) flight has yielded enough data to start presentations and the first manuscript is in preparation. The abstract of the first presentation at IAA April 14, 2011 is listed in the below Bibliography. Work is underway to find additional novel genes that are differentially regulated in microgravity thus providing new targets for treatment of immunosuppression.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Abstracts for Journals and Proceedings	Hughes-Fulford M, Spurlock S, Chang TT, Candelario TT. "Loss of Mouse T-cell Immune Response in Astromice Flown on STS-131." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Abstracts for Journals and Proceedings	Hughes-Fulford M, Boonyaratakanakornkit J, Chang T, Li C-F. "Spaceflight alters expression of microRNA during T cell activation." Immunology 2011. 98th Annual Meeting of the American Association of Immunologists, San Francisco, CA, May 13-17, 2011. Journal of Immunology. 2011 Apr;186(1 Meeting Abstracts):109.16. , Apr-2011
Articles in Peer-reviewed Journals	Hughes-Fulford M, Li CF. "The role of FGF-2 and BMP-2 in regulation of gene induction, cell proliferation and mineralization." J Orthop Surg Res. 2011 Feb 9;6(1):8. PMID: 21306641 , Feb-2011
Articles in Peer-reviewed Journals	Chang TT, Hughes-Fulford M. "Monolayer and spheroid culture of human liver hepatocellular carcinoma cell line cells demonstrate distinct global gene expression patterns and functional phenotypes." Tissue Eng Part A. 2009 Mar;15(3):559-67. PMID: 18724832 , Mar-2009